Plant Disease Prediction

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Agenda

- **Why this matters** (problem & goal)
- **Data & approach** (in one picture)
- **Results at a glance** (CNN, ResNet50, EfficientNetB4, ViT)
- **What it means** (business impact + simple web flow)
- **Recommendation & next steps**



Why This Matters

Effect

Crop losses from late diagnosis are costly (yield ↓, chemicals ↑, time ↑)



Limitation

Manual checks are inconsistent and don’t scale across larger fields

Goal

Fast, consistent triage from a simple leaf photo

Problem & Objectives

Problem:

- Farmers need a quick, reliable way to identify leaf diseases without always relying on experts.

Objectives

- Build an image-based classifier that identifies disease class from a leaf photo.
- Compare 4 model families on accuracy, stability, and speed.
- Recommend a deployment-ready option.

Data at Glance

- **Source:** “PlantVillage dataset (Kaggle)”
- **Classes:** 38 (disease/healthy) classes across 14 crops
- **Images:** ~53k RGB leaf photos





Approach

Pipeline:

- Photo → Preprocess (resize 224×224, normalize) → Model → Predicted label

Design choices:

- Standardize inputs (224×224)
- Simple augmentations (flip/rotate/brightness)
- train/val split (80/20)

Model Lineup

- **Baseline CNN:** Our control; simple, fast, strong reference point
- **ResNet50:** Reliable classic with skip connections; can be overfit without regularization
- **EfficientNetB4:** Great accuracy–efficiency via compound scaling
- **ViT (Vision Transformer):** Transformer-based; very strong with transfer learning and bigger datasets
- For ResNet50, EfficientNetB4 and ViT, we use transfer learning and save the best checkpoint

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Results at a Glance

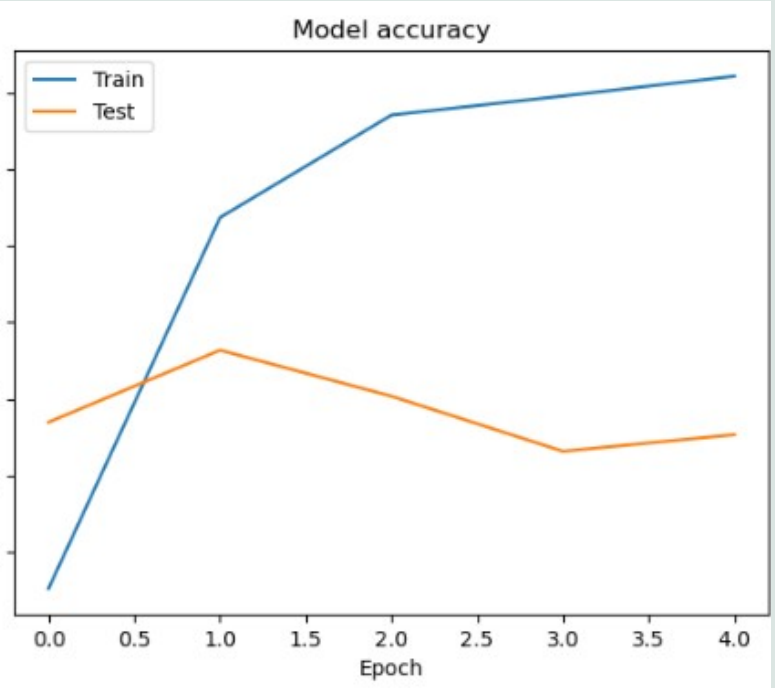
Model	*Val/Test Acc (best)*	*Model size*	*Quick Take*
Baseline CNN	0.89 (early peak; falls to 0.86)	Small	Learns fast then **overfits** after 1–2 epochs.
ResNet50	0.94	Medium	Strong with staged fine-tuning; Top-3 ≈ 1.00
EfficientNetB4 (best)	0.975 (initial TL 0.946)	Medium	Best **accuracy–stability** tradeoff for v1
ViT	0.993	Large	**Highest ceiling; very uniform per-class results.**

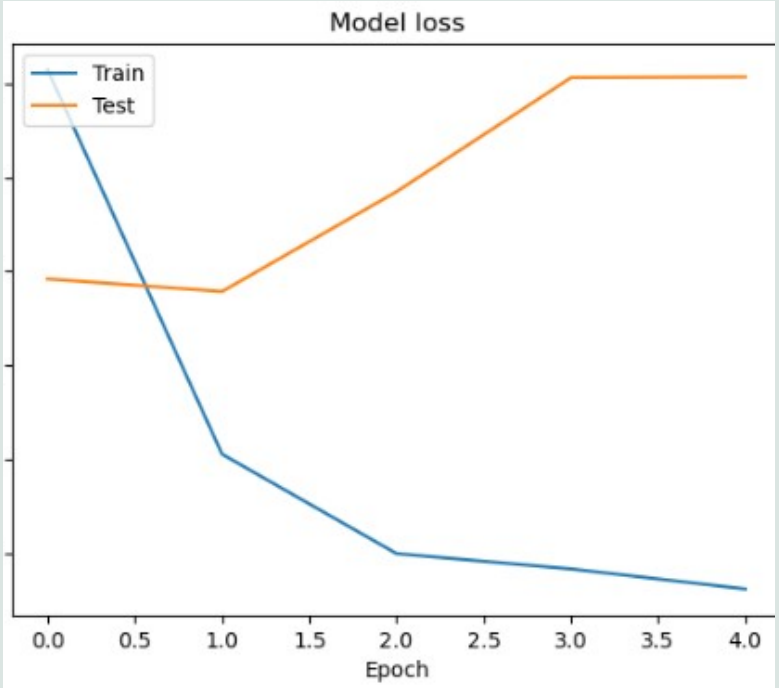
Training Setup

- Input: 224×224 RGB; normalize to [0,1]
- Batch size - 32; optimizer Adam; cross-entropy loss
- Transfer learning for ResNet50 / EfficientNetB4 / ViT
- Early stopping & checkpointing; 80/20 train/validity split

Training Curves

BASIC CNN

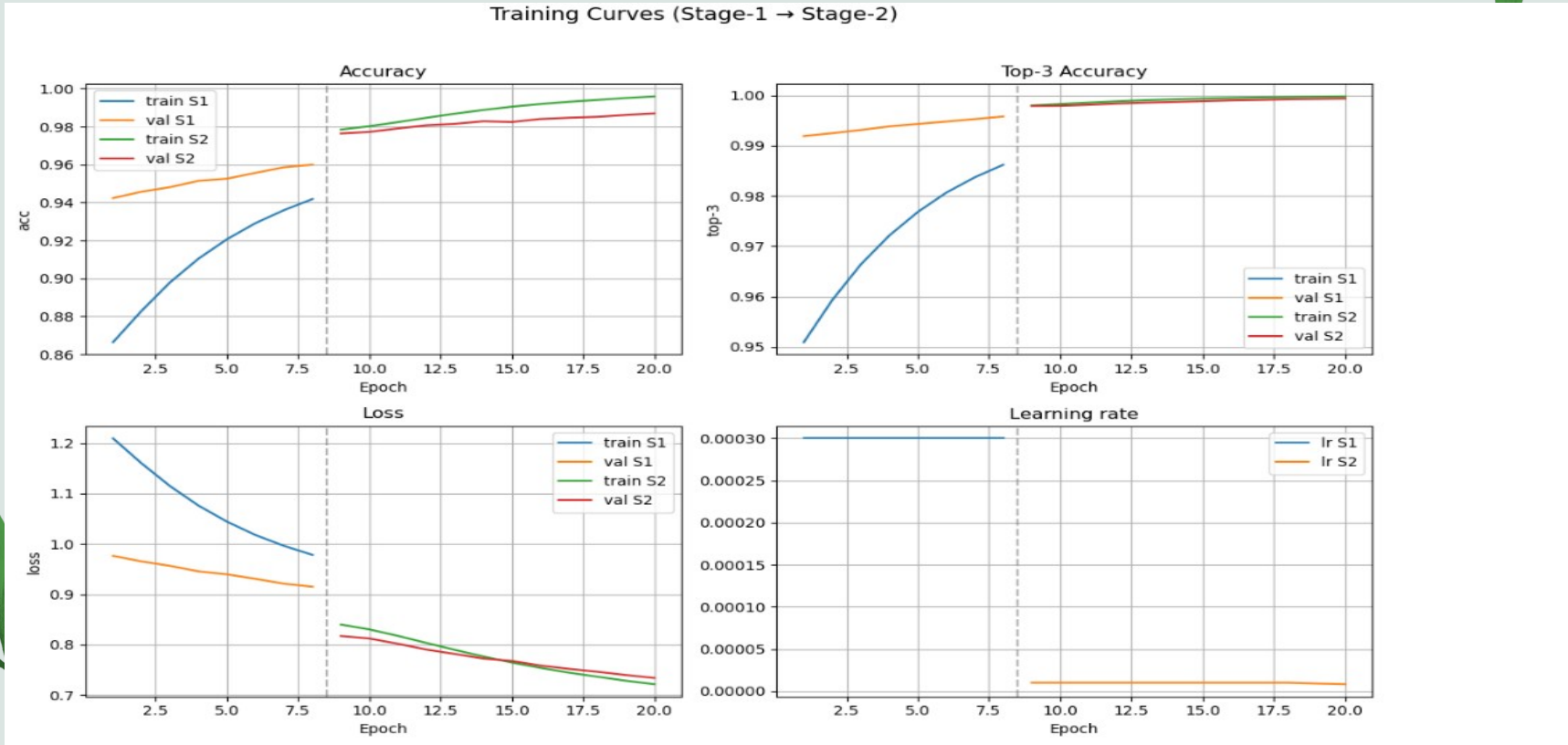




Learns fast then overfits, validation accuracy drops after epoch 1–2.

Training Curves

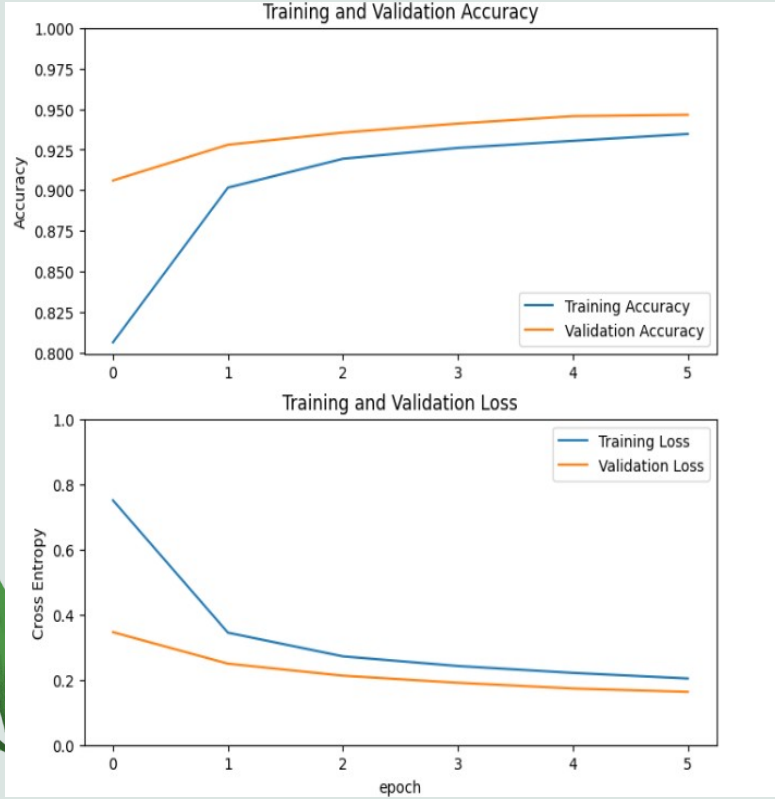
RESNET50



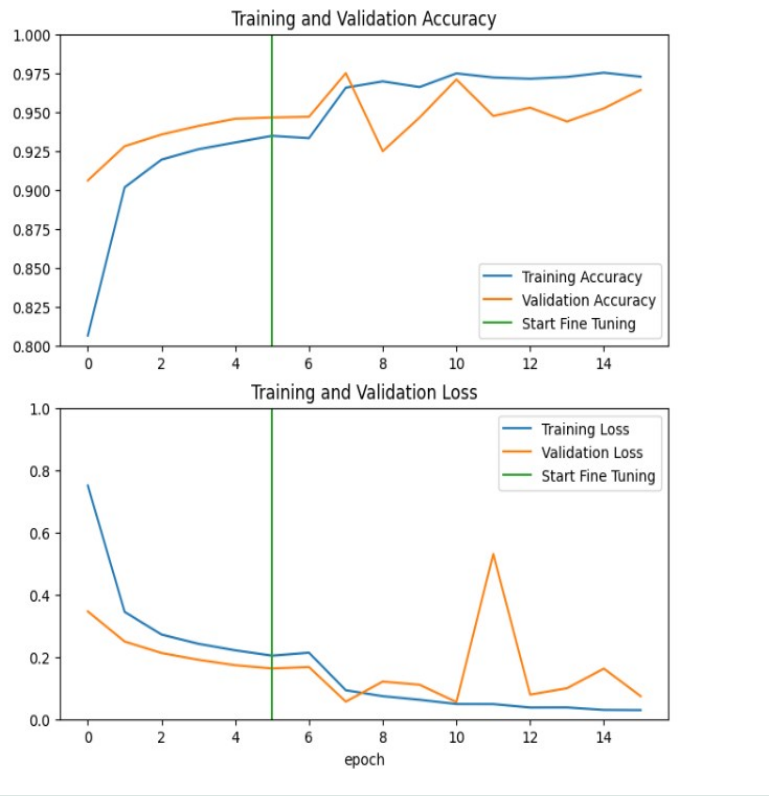
ResNet50: “Stage-2 fine-tuning lifts val accuracy toward ~99%; top-3 ~100%.”

Training Curves

EfficientnetB4

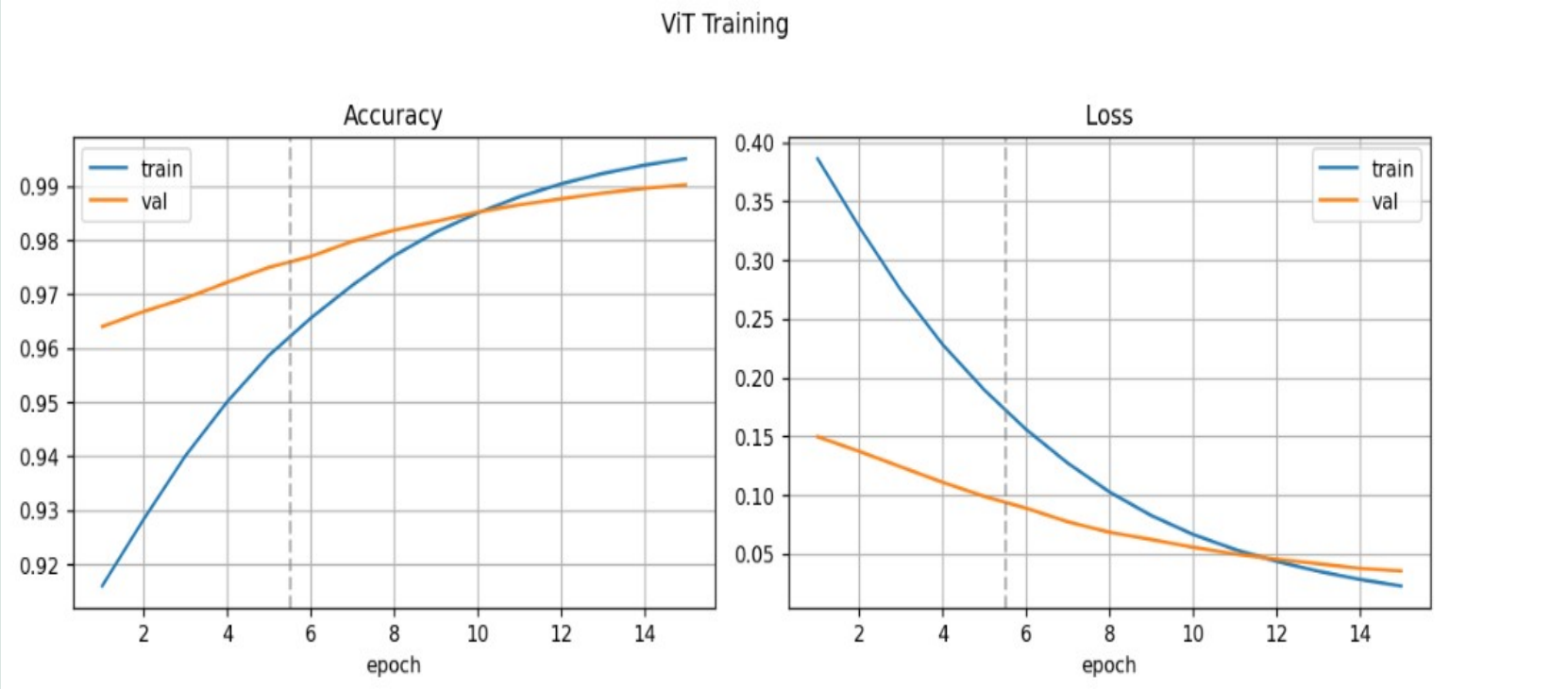


Fine-Tuned EfficientnetB4



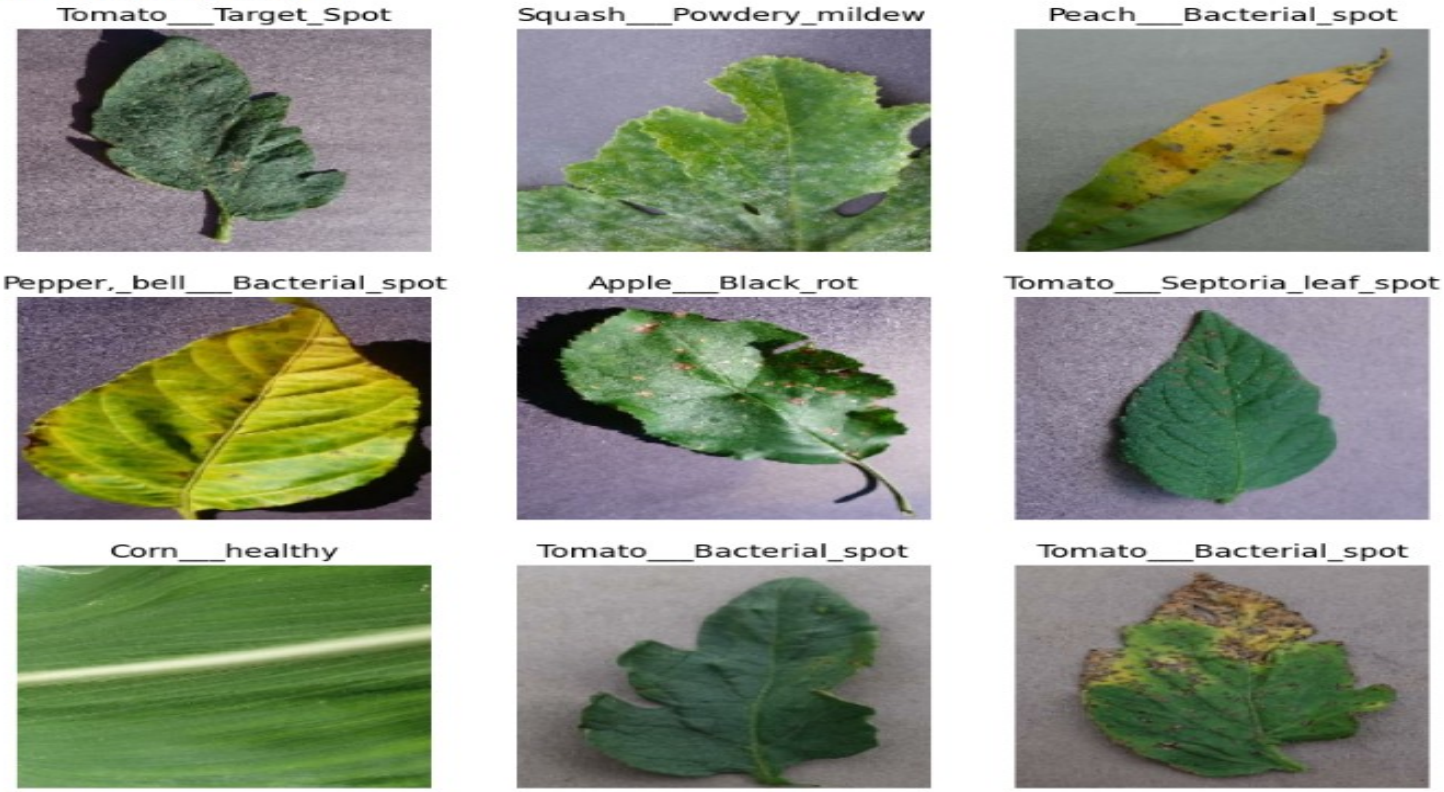
- **EfficientNetB4 (initial):** “Stable mid-90s with minimal tuning.”
- **EfficientNetB4 (fine-tuned):** “Peaks ~97.1%; choose best epoch; LR spikes.”

Training Curves

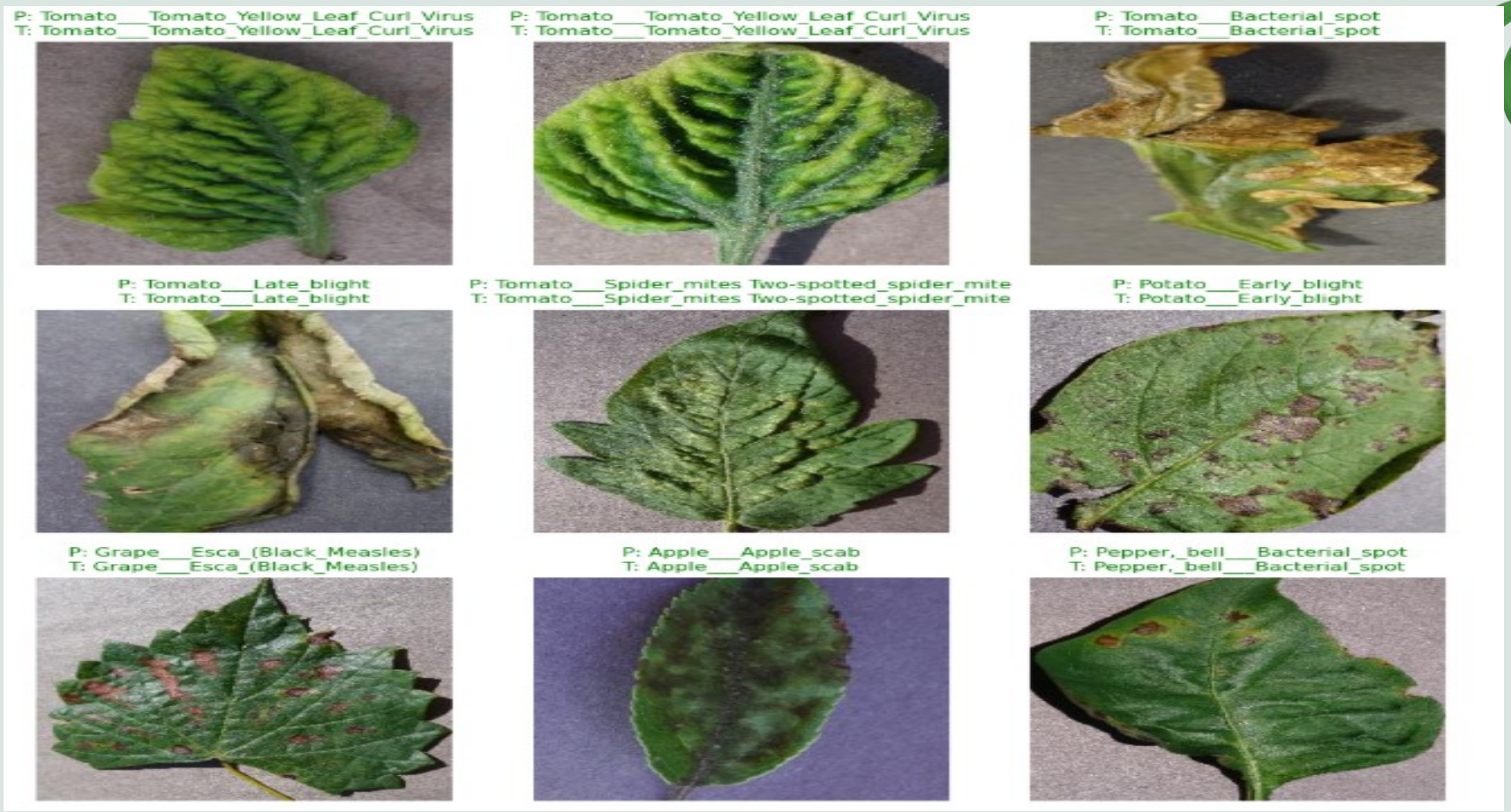


EfficientNetB4 Prediction

```
Labels:
 [35 26 17 19  1 33 11 29 29 13 26  9 28 25 35  0  0 16 24  9 29 35  9 12
 14  6  2 33 21 16  1 14]
```



ViT PREDICTIONS



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Error Analysis & Explainability

ViT classification report :
Accuracy - 0.993; Macro F1 - 0.991.

EfficientNetB4 classification report:
Accuracy - 0.974; Macro F1 - 0.986

Insights:

- Both models are **balanced across classes**; most F1 scores ≥0.98.
- The few dips (≈0.91–0.95) occur on **look-alike lesions** or challenging lighting.
- This aligns with agronomy intuition; guides **targeted augmentation** and future field-data collection.



Business Impact

What these results mean in the field

- **Earlier detection:**
  Lower crop loss. With EfficientNetB4 (97% validation accuracy), we catch issues earlier so fewer plants progress to severe stages.

- **Faster decisions:**
  A phone scan takes 1–2 seconds vs manual checks that can take minutes per patch → scalable across large fields.

- **Consistency across locations:**
  Confidence thresholds + simple UI standardize decisions and reduce variability between scouts.

- **Expert time where it matters:**
  Low-confidence cases are escalated to an agronomist; high-confidence cases are handled on the spot.



- **Operational traceability:**
  Every prediction (image + label + confidence) is logged for QA and continuous improvement.

- **Mini ROI :**

  Potential savings/season = Acres × Affected% × Yield/acre × Price × Improvement %.

- ***Example:***
  500 acres × 20% × 3 tons/acre × $200/ton × 10% = **$6,000** avoided loss.

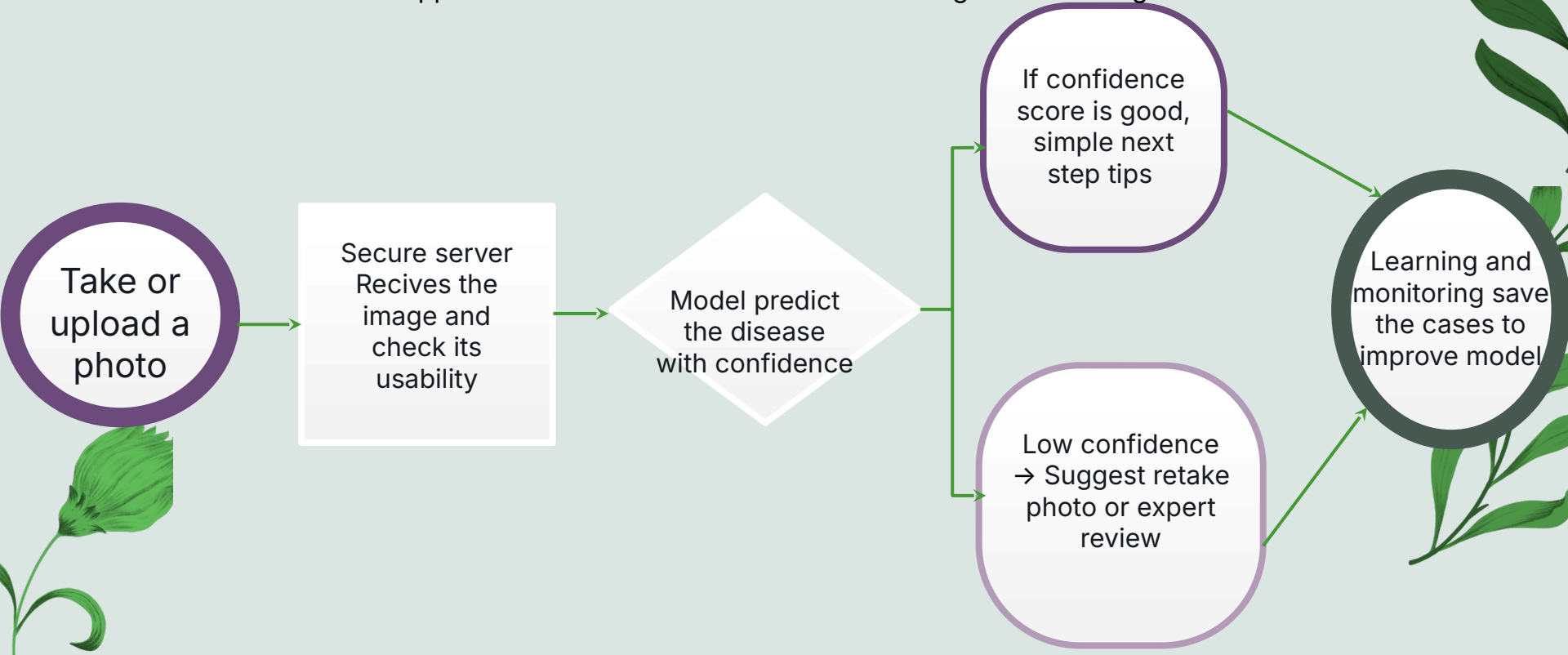
- **KPIs:**

  - Recall on high-impact classes ≥ 97%; False-negative rate ≤ 3%
  - Avg triage time ≤ 5s per leaf

Web Deployment Flow

Flowchart layout

1) Browser → 2) Web App → 3) API → 4) Model Service → 5) Storage & Monitoring



Risks & Safeguards

- **Data drift:** new varieties, lighting, field conditions
  Mitigation: periodic retraining with field photos
- **Bias to lab images:** Distribution shift
  Mitigation: collect field data; augment with realistic noise/angles
- **Confidence misuse:** overtrust low-confidence outputs
  Mitigation: show confidence + ‘re-shoot’ prompt; escalation path

Recommendation & Next Steps

- **Recommendation:**
  EfficientNetB4 as production candidate.

- **R&D track:**
  Keep **ViT** for larger/field dataset.

Next Possible Steps:

- Collect 500–1,000 field images
- Calibrate thresholds; add confidence UI
- Pilot with 1–2 sites; monitor errors; retrain once

THANK YOU

